Deep-learning Lab Project Report
Actuated Metamaterial

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***Abstract*—The behavior of metamaterials is determined by their internal geometry rather than their bulk material. Adding an active phase enables their mechanical response to be tuned on demand. This work presents a new type of actuated metamaterial consisting of a hollow thermoplastic polyurethane (TPU) beam filled with a magnetorheological fluid (MRF) and placed on a permanent magnet. When a magnetic field is applied, the particles inside the MRF align along the field lines, stiffening the fluid and effectively turning the beam into a structure whose stiffness can be controlled magnetically. The model is built in COMSOL Multiphysics by coupling the Solid Mechanics and Magnetic Fields modules. The data obtained from these simulations is then used to train a set of neural networks that act as a digital shadow of the metamaterial, reproducing its behavior without the need for any further FEM analysis. Alongside a direct model, which predicts stress based on the magnetic field, geometry, and strain, and an inverse model, which recovers a geometric parameter based on the target response, cascaded networks are also trained to estimate the internal magnetic field distribution along the beam axis. The surrogate accurately captures the nonlinear response of the beam at a fraction of the cost of the finite element simulations.**

***Index Terms*—Metamaterial, Magnetorehological fluid, TPU, Buckling, COMSOL Multiphysics, Machine Learning, Digital Shadow**

I. Introduction

THIS report presents the study of an actuated metamaterial, based on a simple beam filled with a magnetorehological fluid (MRF).

The final aim is to create a *digital shadow* leveraging on machine learning (ML), to recreate the behavior of the metamaterial having to perform any finite element (FE) simulation. An MRF contains in fact magnetic particles, so that its fluid and mechanical properties change when a magnetic field is applied. Applying a magnetic field aligns the magnetic particles along the field lines and forms ferromagnetic chains arranged in specular, blade-like structures that provide additional stiffness to the structure. Therefore, the MRF can be considered a region of variable stiffness depending on the magnetic field.

The model starts from the analysis of the constitutive behavior of the MRF, then the design and simulation of the metamaterial are assessed, moving from purely mechanical simulations to inserting a magnet in order to exploit the magnetorehological properties of the material. The analysis is developed in small deformation and in the linear elastic regime, even though many metamaterial structures manifest buckling phenomena. Finally, a comparison among different models is provided.

Lastly, a ML approach is proposed, first the dataset creation based on the previous model is described, then the neural network execution is reported.

II. MR Fluid

As briefly outlined in the Introduction, a magnetorehological fluid is a non-Newtonian fluid consisting of ferromagnetic microparticles randomly suspended in nonmagnetic liquid, that quickly changes viscosity in response to an applied magnetic field [5].

For the development of the proposed actuated metamaterial, the following MRF is picked from *ARUS MR TECH* company: *AMT - RHEOTEC*$^+$ [4], a silicon-based MRF.

While the manufacturer's datasheet provides a comprehensive set of physical and rheological parameters, the properties most relevant to the mechanical modeling used in this article are summarized in the figures, namely:

- Fig. 1: yield shear stress τ_y [kPa] vs magnetic field intensity H [kA/m] curve;
- Fig. 2: magnetic flux density B [T] vs magnetic field intensity H [kA/m] curve.

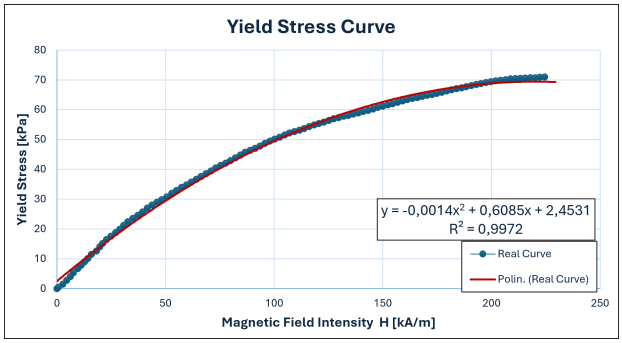


Fig. 1. MR Fluid's yield stress curve from vendor datasheet (blue) and it's own polynomial interpolation (red).

A. MRF model creation

The literature has shown that a good model for such a material must be chosen depending on the region of deformation in which you are interested. In particular, [2] has shown that three regions exist: a linear region (region I) for small deformations, a second region (region II) where compressive stress remains constant, and a third region (region III) where non-linear behavior is exhibited. The chosen model is expected to be effective in region I, where the well-known Bingham model applies. Furthermore, the quasi-static process assumption is made, leading to the

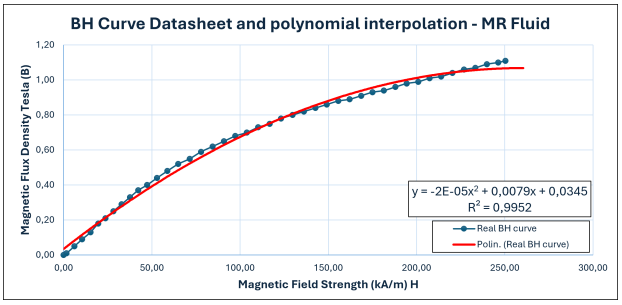


Fig. 2. MRF's $B-H$ curve (blue) from vendor datasheet, with its polynomial interpolation (red).

viscosity term being neglected in the Bingham model.

An experimental approach [2] proposes the following relation between compressive stress and magnetic field flux:

$$\sigma_{c,y}(H) = k \cdot B(H) + \sigma_{\min}. \tag{1}$$

where $\sigma_{c,y}$ is the maximum value of compressive stress in the viscoelastic region I; k and $\sigma_{\min}$ are model parameters to be tuned with experimental tests on the MRF.
A first-approximation for the estimation of the Young's modulus is the following expression:

$$E_{eq}(H) = \frac{\sigma_{c,y}(H)}{\epsilon_{c,y}} = \frac{k \cdot B(H) + \sigma_{\min}}{\epsilon_{c,y}}. \tag{2}$$

where $\epsilon_{c,y}$ is the yield strain under compression.

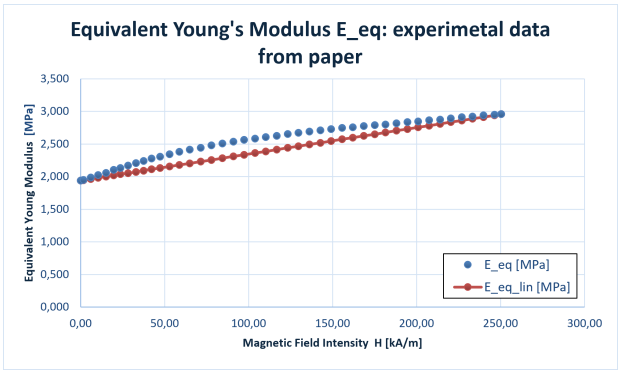


Fig. 3. Effective Young's modulus with datasheet data [4] and experimental data in paper [2] (blue), and the linear interpolation (red).

Therefore, depending on the type of MRF a curve may be found from experimental data as shown in Fig. 3, from which a linear regression model may also be obtained.

Nevertheless, in this discussion a different approach is followed based purely on theoretical knowledge and datasheet curves. As the model is assumed to work in the linear regime, a linear fit of the quantity in datasheet is proposed:

$$\tau_y(H) = m_\tau \cdot H. \tag{3}$$

which it is supposedly valid up to a magnetic field magnitude of 100 kA/m. The slope is computed by taking the following intermediate points:

$$m_\tau = \frac{27\ \text{kPa} - 12\ \text{kPa}}{120\ \text{kA/m} - 40\ \text{kA/m}} = \frac{15\ \text{kPa}}{80\ \text{kA/m}} = \\ = 0.1875\ \text{kPa / (kA/m)}. \tag{4}$$

In literature [1] it is shown that γ is not changing much, hence it is set equal to 0.003. As a consequence:

$$G_{eq}(H) = \frac{\tau_{c,y}(H)}{\gamma_{c,y}}. \tag{5}$$

Assuming isotropic material, an equivalent Young's modulus is derived:

$$E_{eq}(H) = 2 \cdot G(H) \cdot (1 + \nu) = 188.8\ \text{kPa/T} \cdot H \tag{6}$$

Finally, the constitutive law for the MRF is:

$$E_{\text{eq-lin}}(H) = (188.8 \cdot H + q_\tau)\ \text{kPa}. \tag{7}$$

where $q_\tau = 1$ Pa to avoid a zero Young's modulus when no magnetic flux density is applied.

The displayed approach has to be verified experimentally as stated above, however no experimental studies have been done in this work and the (7) is assumed to be hold true.

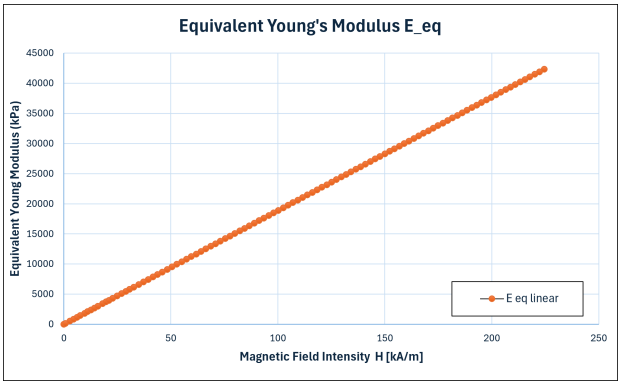


Fig. 4. Effective Young's modulus adopted in the numerical simulation.

III. Thermoplastic Polyurethane

The thermoplastic polyurethane (TPU) parameters are set as follows:

$$\rho = 1200\ \text{kg/m}^3 \tag{8}$$
$$\mu_r = 1 \tag{9}$$
$$E_{TPU} = 7.5\ \text{MPa} \tag{10}$$
$$\nu = 0.45 \tag{11}$$
$$\epsilon_r = 1 \tag{12}$$

where ρ is the TPU density, μ_r is its relative magnetic permeability, E_{TPU} is the Young's modulus, ν is the Poisson's coefficient, ϵ_r is the relative electric permittivity.

IV. Model creation

A. Geometry model definition

The metamaterial beam is geometrically defined as shown in Fig. 5. TPU whose thickness is TPU_{th} constitutes the cylindrical beam's external wall with maximum radius TPU_R. The hollow interior of the cylinder of length MR_L and of radius MR_R is filled with the MRF. A permanent magnet of length MT_L is placed beneath the beam.
For the final parametrization and the parametric description of the quantities see Section IV-C.

B. Buckling analysis

Over time, the beam undergoes a compressive imposed displacement, reaching a maximum displacement of u. For the purposes of our study, the model must remain linear, so a displacement equal to 10% of the initial length is imposed.
Moreover, an initial buckling analysis is performed. This analysis helps to understand the ratio of TPU_L to TPU_R for creating the future dataset. This allows to change the dimensions without any major problems or nonlinearities occurring due to buckling.
The cylindrical hollow beam is schematized as being entirely made of TPU, while the internal cavity is filled with MRF. The effective length factor $k = 0.5$, which multiplies the initial length, is imposed as the beam extremes are both constrained to move vertically (i.e. it is a fixed-fixed configuration, see Fig. 5). A, I, E respectively mark the area of the beam of radius TPU_R, the moment of inertia and the TPU beam Young's modulus.
The study leads to consider the following expressions:

$$u = 0.1 \cdot TPU_L, \tag{13}$$

$$P = \frac{E \cdot A}{TPU_L}, \tag{14}$$

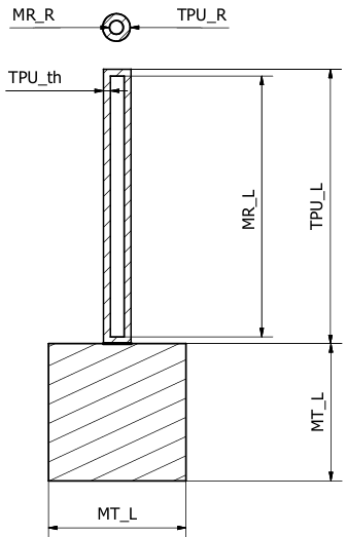


Fig. 5. Geometry of the TPU beam with the magnet at its bottom.

$$P_{cr} = \frac{\pi^2 \cdot E \cdot I}{(2 \cdot TPU_L)^2}, \tag{15}$$

$$I_{\text{circular}} = \frac{\pi \cdot TPU_R^4}{64}. \tag{16}$$

The resulting ratio between TPU_L and TPU_R is showed in Eq. (17):

$$TPU_L = \sqrt{\frac{\pi^2 \cdot TPU_R^2}{4 \cdot k^2 \cdot 0.1}} = 9.95 \cdot TPU_R. \tag{17}$$

C. Final beam parametrization

Making all beam dimensions depend solely on TPU_L implies that the beam can adapt dynamically when this parameter is varied in isolation, preventing buckling:

$$TPU_R = \frac{TPU_L}{9.95} \tag{18}$$

$$TPU_{th} = 0.05 * TPU_L \tag{19}$$

$$MR_L = TPU_L - TPU_{th} \tag{20}$$

$$MR_R = TPU_R - TPU_{th} \tag{21}$$

The magnet has instead a fixed size $MT_L = 20$ mm.

V. Setup of the multiphysics simulation

The beam model is built in Comsol Multiphysics™, with the geometrical parameters showed in Section IV-C.
"Solid Mechanics" and "Magnetic fields, No Currents" modules are used for the simulations.

A. Time-dependent compression

The beam is subjected to a compressive load over time, with a maximum imposed displacement u as in Eq. (13) and a time dependent analysis is carried out. The compression is schematized as a function of t_{cruise}, t_{smooth}, t_{tot}, the imposed displacement and the compression velocity v_{comp}, where:

$$t_{\text{tot}} = t_{\text{smooth}} + t_{\text{cruise}}, \tag{22}$$

$$t_{\text{cruise}} = \frac{u}{v_{\text{comp}}}, \tag{23}$$

$$t_{\text{smooth}} = 0.1 \cdot t_{\text{cruise}}. \tag{24}$$

Specifically, t_{smooth} is the time interval for the initial "smooth step", which is set to prevent having infinite acceleration at the initial time instant during the compression, due to the instantaneous values of v_{comp}. Alternatively, t_{smooth} can be seen as the time to reach the constant velocity v_{comp}.
t_{cruise} is the time in which the speed remains constant.
The initial smooth step follows the formulation shown in Eq. (25), which is derived from the kinematic relationship of a point moving with constant acceleration followed by constant speed. In Comsol Multiphysics™ it is assembled as a piecewise function, $pw(t)$:

$$pw(t) = \begin{cases} \frac{v_{\text{comp}} \cdot t^2}{2 \cdot t_{\text{smooth}}}, & \text{if } 0 \le t \le t_{\text{smooth}} \\ \frac{v_{\text{comp}} \cdot (t - t_{\text{smooth}})}{2}, & \text{if } t > t_{\text{smooth}} \end{cases} \tag{25}$$

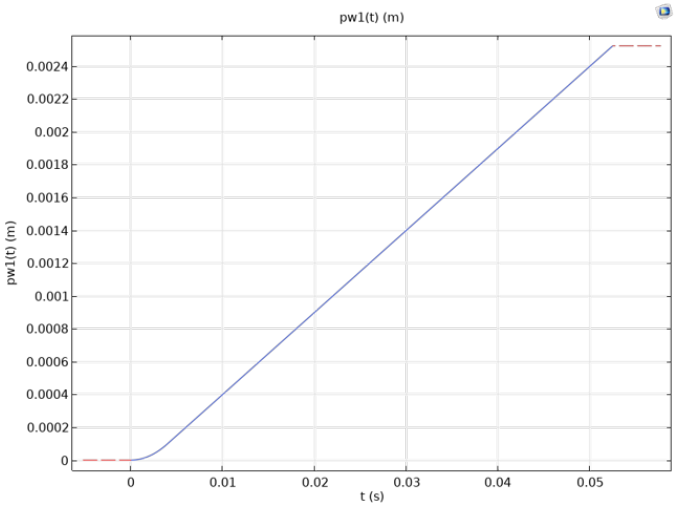


Fig. 6. Piecewise displacement function implemented for the smooth displacement at the start of the analysis.

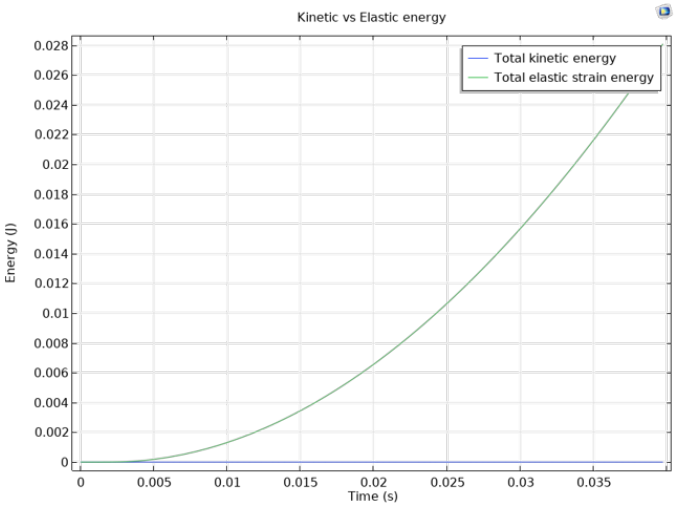


Fig. 7. Kinetic and elastic energy comparison for the selection of the velocity v_{comp}.

The total simulation time, t_{tot}, is discretized in 100 steps. Lastly, it is worth noting that the compression velocity v_{comp} must be carefully set, as a too small value would result in a very long simulation, while a too high value would generate wave propagation and other dynamical effects that violate the hypothesis of a quasi-static process. To select the correct velocity, the kinetic and elastic strain energies are compared so that their values differ by at least an order of magnitude, as shown in Fig. 7. Finally, the imposed value is $v_{\text{comp}} = 50$ mm/s.

B. Material selection

The material are selected coherently with the geometry of the beam: the external walls are made of TPU, whose physical properties are described in Section III, while the internal hollow part is made of the MRF, whose physical properties are collected in Section II.
An air enclosure is required to ensure the correct spatial distribution of the magnetic flux field B. This enclosure is represented by a cylinder with a height of Air_L and with a diameter of equal length. Beam and magnet are placed inside this domain.
The final geometrical representation is showed in Fig. 8 and it is set Air_L=90 mm.

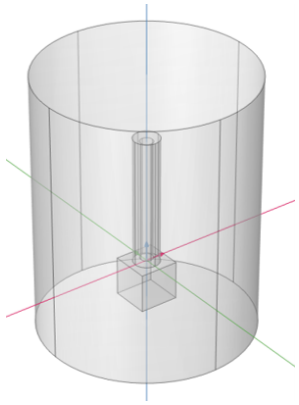


Fig. 8. Model geometry for simulations in Comsol Multiphysics™, including magnet, beam, and air domain (red: X axis, blue: Z axis, green: Y axis).

C. Boundary conditions

To follow the analytical scheme described in Section IV-B, the physical boundary conditions are replicated in the software, by imposing a *Fixed Constraint* at the base and a null *Prescribed Displacement* on the beam top surface along the X and Y directions. Along the vertical Z direction the *Prescribed Displacement* is set in accordance with Eq. (25).

D. Mesh generation

The adopted mesh is *Physics-controlled*, with the element size set on *Finer*.
No further adjustments have been made to the mesh generation. The results obtained using this technique (presented in the next Section) are satisfactory; refining or reducing the overall element size provided no notable improvements.
Furthermore, including the air domain makes implementing a moving mesh formulation at the air boundary essential in order to mitigate mesh distortion and prevent null or negative element Jacobians. This approach inherently accommodates geometric nonlinearities within the numerical computation; however, these remain negligible under the small displacement assumption.

E. Solver configuration

Default software configurations are adopted for the solver setup; specifically, in those models where it is needed, a fully coupled approach is selected, utilizing the *MUMPS direct* solver to compute the model.

VI. Beam simulations

The Comsol Multiphysics™ settings mentioned above, unless explicitly stated otherwise, are used for building different models. This approach deals with two stages of increasing complexity:

- a baseline model considering exclusively a solid mechanics domain under compression;
- a multiphysics model incorporating solid mechanics coupled with magnetic field simulation, and accounting for the permanent magnet, the MRF domain, and its nonlinear $B - H$ curve.

This incremental strategy enables to evaluate the effect of each added layer of complexity on the overall results, while facilitating the systematic validation of the developed models. The quantities of interest are the stress-strain curve throughout the entire time of compression and, where present, the distribution of the norm of the magnetic flux field along the vertical axis of the beam, as this regulates the magnetorehological properties of the MRF.
The nominal stress σ is calculated as in Eq. (26); conversely, the nominal strain ϵ is computed as in Eq. (27). The absolute value is needed because a compression is imposed.

$$\sigma = \frac{|F_{\text{Z-reaction}}|}{A_{\text{0-beam}}} = \frac{|F_{\text{Z-reaction}}|}{\pi \cdot TPU_R^2}. \tag{26}$$

$$\epsilon = \frac{u}{TPU_L}. \tag{27}$$

A. 3D and 2D axisymmetric models

The beam is developed in both 3D and 2D axisymmetric configurations. The 2D version speeds up calculations, as it involves solving only one physical domain with a lower number of nodes. The drawback is that the 2D simulation does not include the buckling phenomenon, as axial symmetry prevents this effect. For this reason, the imposed displacement u is carefully analyzed during the simulation and iteratively compared between the 3D and 2D models. This leads to use Eq. (28) instead of Eq. (13), which allows keeping the beam parametrization defined in Section IV-C.

$$u = 0.05 \cdot TPU_L. \tag{28}$$

This reduction is particularly significant when considering the TPU only in the external wall, as our initial analysis in Section IV-B was taking in consideration a full TPU beam with no internal cavity.
Based on these considerations, it can be shown that the 3D and 2D models have the same behavior in terms of the stress-strain curves, as Fig. 9 shows. These results mean that the 2D model can be used to speed up calculations in future datasets.

B. Solid mechanics models

To understand the behavior of the beam in compression only, the *Solid Mechanics* module is used and three alternatives are explored.

1) Full TPU beam: the first model considers a beam made of TPU only without any cavity, which allows finding the new maximum allowed displacement described by Eq. (28).
The results have been already shown in Fig. 9, where the two curves perfectly match.

2) Model with different material in the beam cavity: the second model tackles the effect of the internal geometry for the future implementation of MRF. Hence, the external walls are made of TPU, while the internal cylinder is defined as a material with a higher Young's modulus, $E_{\text{internal}} = 30$ MPa. As expected, the stress is higher than the previous case, around 0.62 MPa (see Fig. 10), compared to 0.37 MPa (Fig. 9).

3) Solid mechanics model with magnetic field imposed: the third model evaluates the correctness of the MRF response under different magnetic field intensities H, using exclusively the *Solid Mechanics* module. To maintain a purely solid mechanics framework, the MRF is modeled through an effective Young's modulus, see Eq. (7), as a function of the parameter H that is varied during the simulation. As shown in Fig. 11, for high H, the effective Young's modulus of the MRF increases, and the stress is higher at a given strain.

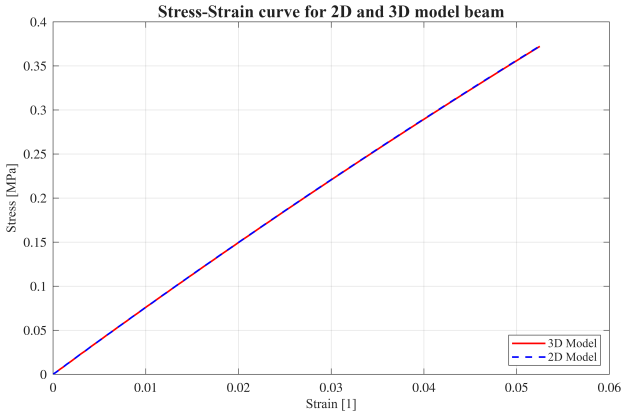


Fig. 9. Stress-strain curve for the 2D and 3D model beam made only of TPU without hollow parts.

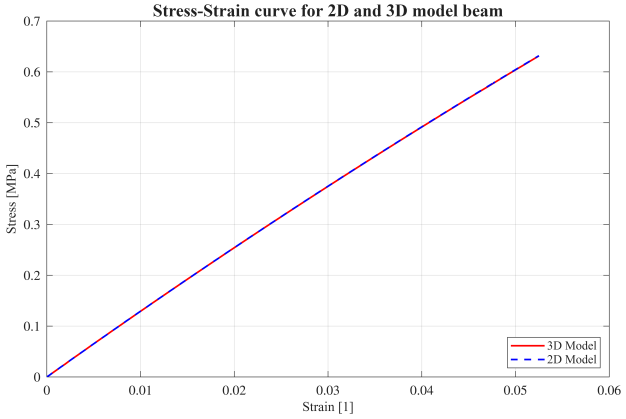


Fig. 10. Stress-strain curve for the 2D and 3D model beam with the internal MRF modeled as a solid material with higher E.

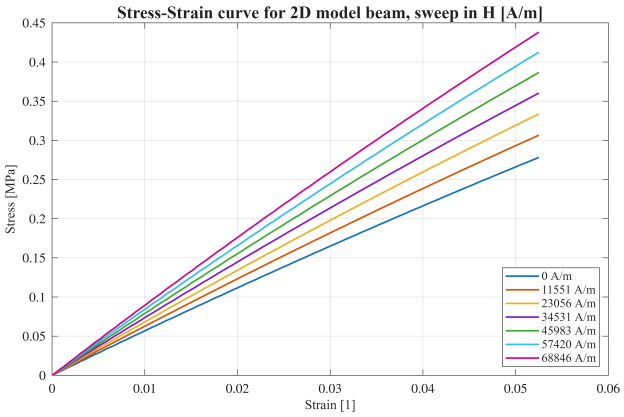


Fig. 11. Stress-strain curve for the 2D model, in the *Solid Mechanics* module, as a function of different H values.

C. Multiphysics model

In the multiphysics model, the magnetic field intensity H is a consequence of the presence of the magnet, whose constant magnetic field flux is B.
Consequently, the *Magnetic Fields, No Currents* module is added to the workspace; as electric current interactions are neglected, computations are significantly sped up. The permanent magnet is implemented by specifying its north and south polarities. Then, the *Magnetization model* is set to *Remanent flux density* to define the specific flux density within the simulation, which generates the magnetic field distribution H shown in Fig. 12.

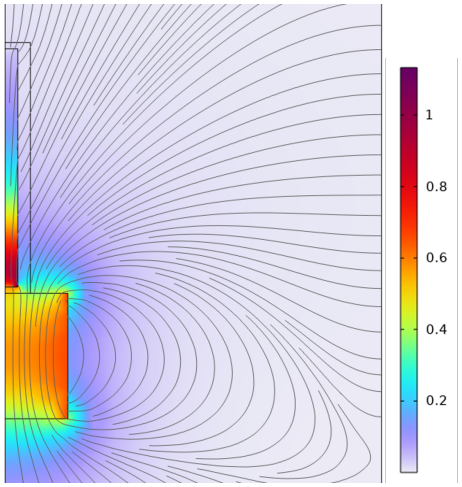


Fig. 12. Norm of the magnetic flux density B, measured in T, for the 2D model.

Two alternative models are compared in this Section.

1) Magnet without $B-H$ curve in MRF model: this model addresses the response of the beam with constant magnetic permeability of the MRF set to $\mu = 6.5$. The obtained stress-strain curve is shown in Fig. 13.
The presence of the magnet alters the magnetic field intensity H distribution, as illustrated in Fig. 14. The plot outlines H against the coordinate z moving along the beam axis, with the origin placed at the point of minimum clearance between the beam and the magnet inside the MRF cavity. The H distribution is coherent with the model representation in Fig. 12: a peak of the H distribution is visible near the magnet, while, as the distance from the magnetic field source increases, a long tail manifests.

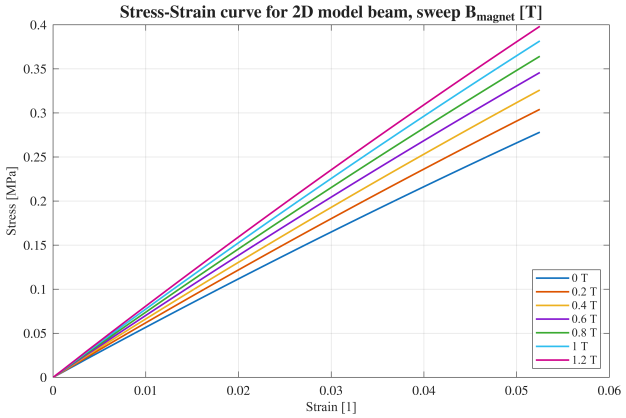


Fig. 13. Stress-strain curve for the 2D model, as a function of different values of the magnet magnetic field flux B, without considering the $B-H$ curve.

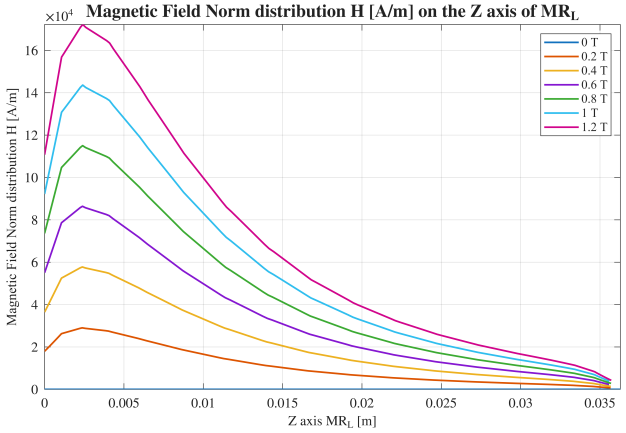


Fig. 14. Norm of the magnetic field H inside the MRF along the z coordinate along the beam axis, without considering the $B-H$ curve.

2) Magnet and B-H curve in the MRF model: the $B-H$ curve is included in the model, thus a nonlinear magnetic permeability μ due to the MRF material is considered.
The $B-H$ curve shown in Fig. 2 is uploaded in the material properties of the software; the consequent results, in terms of stress-strain curves, are shown in Fig. 15, and, in terms of the distribution H, in Fig. 16. The difference from the previous step is not really appreciable, as the next comparison will make clearer.

Comparison between models: the first comparison is made between the standalone *Solid Mechanics* model (SM) and the full multiphysics simulation (MG). First, the H distribution along the MRF vertical axis (plotted in Fig. 14) is extracted for each value of B in the parametric sweep, and its mean value is computed for each curve. These mean values are then applied as inputs to a purely *Solid Mechanics* model. To maintain a consistent and streamlined presentation, these

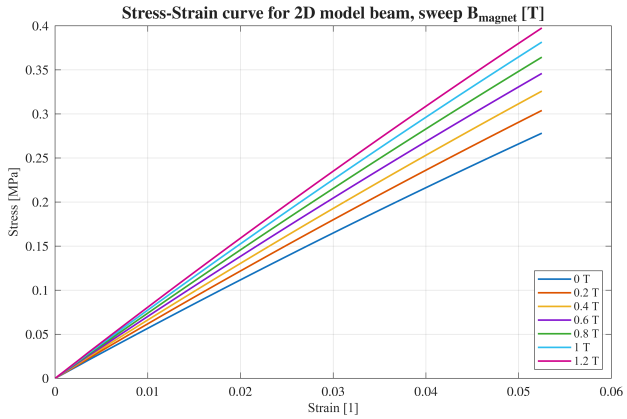


Fig. 15. Stress-strain curve for the 2D model, as a function of B generated by the magnet, when the $B-H$ curve is included in Comsol Multiphysics™ material properties.

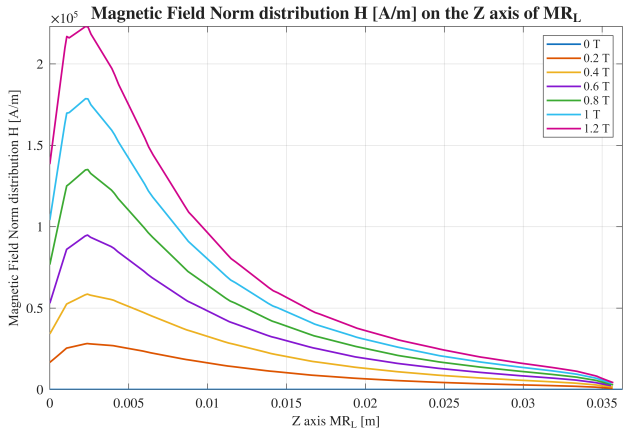


Fig. 16. Norm of the magnetic field H, inside MRF, along the z coordinate moving along the beam axis, when the $B-H$ curve is included in Comsol Multiphysics™ material properties.

simulation values correspond to those already discussed in the previous Section. Consequently, the final results are obtained by superimposing the curves from Fig. 11 and Fig. 13 into a single plot, as shown in Fig. 17. The second comparison is performed to evaluate the impact of implementing the B-H curve within the MR fluid material definition. Following the same methodology as before, the H distribution along the beam axis is extracted and the resulting stress-strain curves are overlaid across the same parametric sweep of B. The magnetic field distributions from the previous models are combined into a single plot in Fig. 18. Incorporating the B-H curve leads to higher peaks compared to the baseline magnet model. Although this difference is minor, i.e. on the order of a few kA/m, and introduces only slight variations in the MRF Young's modulus, the trend is consistently reflected in the stress-strain curves shown in Fig. 19. To clearly observe these subtle deviations, a close-up view is provided in Fig. 20. It is clear that while the small variations introduced by the B-H curve do not significantly affect the global stress-strain behavior, the presence of this nonlinear relation still exerts a substantial impact on the spatial distribution of the H-field as the imposed magnetic flux B increases.

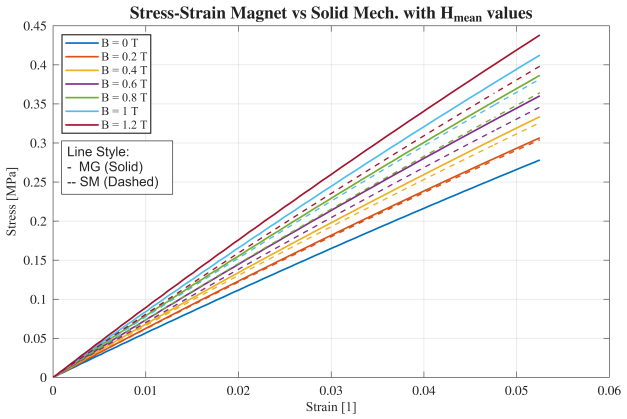


Fig. 17. Stress-strain curve for the multiphysics (MG) model as a function of B, after extracting the H distribution, calculating the mean, and simulating into the *Solid Mechanics* (SM) model.

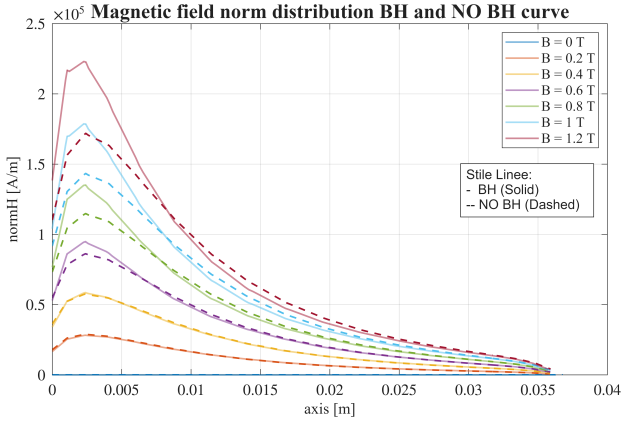


Fig. 18. Norm of the magnetic field intensity H along the beam axis for the multiphysics model, with and without $B-H$ curve implementation.

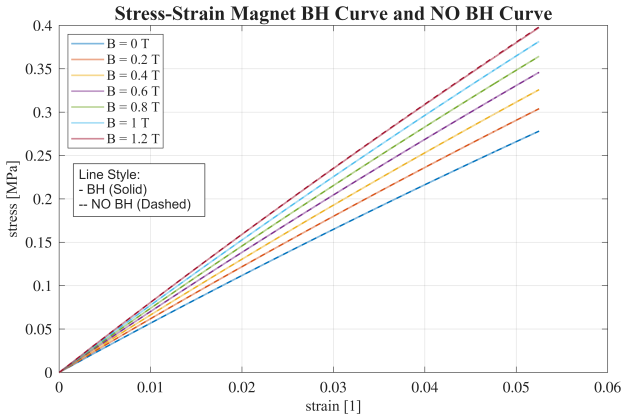


Fig. 19. Stress-strain curves for the multiphysics model, with and without $B-H$ curve implementation.

D. Decoupled simulation

The nonlinear magnetic field distribution $H(z)$, extracted along the beam axis at the initial simulation time from the multiphysics model in Section VI-C.2, is used here as an input to define the Young's modulus of the MR elastomer in a solid mechanics simulation. The results show a decrease in the stiffness when the initial magnetic field intensity is

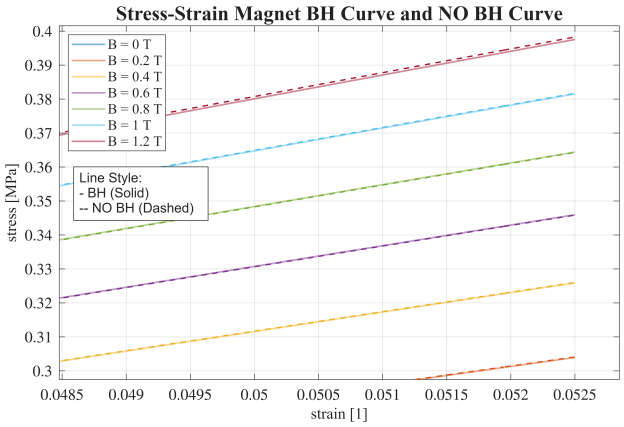


Fig. 20. Magnification of the stress-strain curve in Fig. 19 for the multiphysics model, with and without $B-H$ curve implementation.

applied. Furthermore, unlike the multiphysics approach, the solid mechanics simulation does not account for the localized increase in stiffness caused by the concentration of the B-field during compression, effect which is clearly visible at higher magnetic flux densities.

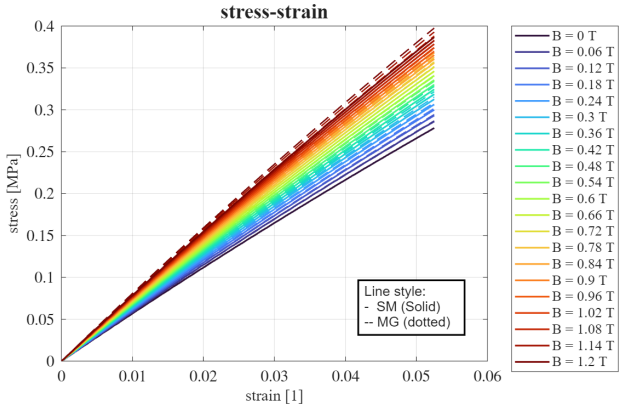


Fig. 21. Stress-strain curve comparison between the solid mechanics (SM) simulation and the multiphysics (MG) model incorporating a permanent magnet.

VII. Machine Learning Implementation

Accurately predicting the mechanical response of metamaterial structures remains a challenging task as the relationship between the geometric parameters, applied strain, and resulting stress can be highly nonlinear and often requires expensive FE simulations or experimental characterization. These approaches are highly accurate but may entail high computational costs. To address this limitation, data-driven techniques have been proven as an effective alternative and in particular, *Artificial Neural Networks (ANNs)* [13] can approximate nonlinear mappings directly from the data, achieving fast predictions once trained.

Inspired by biological neural structures, ANNs are computational models composed of interconnected processing units called neurons (or nodes). These networks learn to recognize underlying relationships in the dataset through an iterative training process, effectively functioning as universal function approximators.

A single artificial neuron processes information by taking a vector of inputs $\mathbf{x}$, multiplying each by a corresponding weight $\mathbf{w}$, adding a bias term b, and passing the result through a non-linear activation function Φ:

$$y = \Phi(\sum_i w_i x_i + b). \tag{29}$$

The inclusion of non-linear activation functions grants the network the mathematical flexibility to capture highly complex non-linear relationships that traditional linear regression models fail to resolve.

When the individual neurons are structured into a forward propagating architecture, they form a *Multilayer Perceptron (MLP)*, a foundational class of *Feedforward Neural Networks*. An MLP consists of three distinct layer types: an *Input Layer* that receives the feature variables, one or multiple *Hidden Layers* where feature extraction and non-linear mapping occur, and an *Output Layer* which yields the network predictions, usually without the need of an activation function.

During training, data is passed forward through the network to generate a prediction (forward propagation). The error between the prediction and the true target value is evaluated using a loss function, and the gradient of this loss is computed via backpropagation. An optimization algorithm then iteratively updates these parameters to minimize error. Once fully trained, the MLP bypasses the need to solve complex differential equations or boundary-value problems in real time.

In this work, Multilayer Perceptrons are used as the surrogate constitutive model for predicting the mechanical response or geometry of the Metamaterial beam. The models were developed using Python and the Pytorch library, which provided flexibility required for the network architecture and training process.

A. Dataset creation

The datasets are created by running the simulation from Comsol Multiphysics™ using the complete model described in Section VI-C2. To satisfy the buckling constraints, the following search space has been considered: beam length $TPU_L \in [20, 40]$ mm, here on marked as L with a step of 1 mm, hence $M = 20$ steps; magnetic flux density $B \in [0, 1.2]$ T in $N = 20$ steps. All possible combinations are performed, yielding a total of 400 simulations. However, the particular simulation for which $B = 0$ is removed being the less interesting case of a simple beam without a magnetic field interaction. Moreover, time spans in $t \in [0, t_{\max}]$ for $N_t = 101$ points, $t_{\max}$ being the time to reach the maximum displacement imposed. For each MLP, specifically suitable datasets S are considered:

- *Direct design MLP*: S is taken as

$$\mathcal{S} = \left\{ \left(B_i, L_i, \{\epsilon_{i,j}, \sigma_{i,j}\}_{j=1}^{N_t} \right) \right\}_{i=1}^{N_{sim}} \tag{30}$$

where $N_{sim} = N \times M$, $B_i \in \mathbb{R}$ is the magnetic flux density imposed by the magnet, L_i is the TPU_L length of the beam considered, $\epsilon_j \in \mathbb{R}$ is the nominal strain,

$\sigma_j \in \mathbb{R}$ is the nominal stress as a consequence of the strain ϵ_j. For a given simulation, i.e. a pair of (B_i, L_i), a stress-strain curve of N_t points is obtained.

- *Inverse design MLP*: S is taken as

$$\mathcal{S} = \{(B_i, L_i, k_i)\}_{i=1}^{N_{sim}} \tag{31}$$

where k is the mean stiffness computed from the $\sigma - \epsilon$ curve for each (B_i, L_i) pair.

- $B(x)$ *and* $H(x)$ *MLP*: S is taken as

$$\mathcal{S} = \{(B_i, L_i, \mathbf{x}_i, \mathbf{B}_i(\mathbf{x}_i), \mathbf{H}_i(\mathbf{x}_i, t))\}_{i=1}^{N_{sim}} \tag{32}$$

where $\mathbf{x}_i \in \mathbb{R}^I$ is the vector of coordinates along the beam axis. I is changing from simulation to simulation due to the different meshes being adjusted for different L_i. $\mathbf{H}_i(\mathbf{x}_i) \in \mathbb{R}^I$ is the magnetic field strength computed along the beam axis coordinates $\mathbf{x}_i$.

- $\Delta H(x)$ *MLP*:
S is taken as

$$\mathcal{S} = \left\{ \left(B_i, L_i, \{\epsilon_{i,j}\}_{j=1}^{N_t}, \mathbf{H}_i(\mathbf{x}_i, t_0), \Delta\mathbf{H}_i(\mathbf{x}_i \right) \right\}_{i=1}^{N_{sim}} \tag{33}$$

where $\Delta\mathbf{H}_i(\mathbf{x}_i)$ is computed as difference of previous and next values of the $\mathbf{H}(\mathbf{x})$ curve at different time steps as a response to the applied $\epsilon_{i,j}$, $\mathbf{H}_i(\mathbf{x}_i, t_0)$ is the initial magnetic field strength curve along the beam axis.

B. Neural network for direct design

The Direct MLP model aims to predict the stress in different physical and magnetic configurations:

$$\sigma = f(B, L, \epsilon). \tag{34}$$

The dataset considered is the S as defined in Section VII-A.

The model implements a feedforward neural network, including an input layer, three hidden layers and an output layer. The input layer contains three neurons, corresponding to the input parameter variables as $B \in \mathbb{R}, L \in \mathbb{R}, \epsilon \in \mathbb{R}$. Each hidden layer contains eight neurons with a ReLU activation function. The output layer has only one neuron, which corresponds to the stress parameter $\sigma \in \mathbb{R}$.

For network training, the Mean Squared Error (MSE) was adopted as metric to measure the difference between the predicted stress and the reference stress values obtained from high-fidelity simulations.

The learning rate $\eta = 10^{-3}$ was selected to ensure a balance between convergence speed and training stability. The Adam optimizer was employed to adaptively adjust learning rates for each parameter based on the first-order (mean) and second-order (variance) gradient estimates. The model was trained for 100 epochs, where one epoch corresponds to a full pass over the training dataset.

After training, the predictive performance of the neural network was assessed using the training, validation, and test datasets. To evaluate this performance, two primary metrics were chosen, namely the Root Mean Square Error (RMSE), which quantifies the magnitude of prediction errors, and the Coefficient of Determination (R^2), which indicates how well the model explains the variance in the target data.

TABLE I
PERFORMANCE OF THE DIRECT MODEL.

Dataset	RMSE	R^2
Training	0.00342	1.000
Validation	0.00346	1.000
Test	0.00346	1.000

The model achieved low prediction errors across all datasets, with similar RMSE values for the training, validation and test sets. The comparable performance indicates that the model generalizes well and does not exhibit significant overfitting.

From Fig. 23 it can be observed that both curves decrease steadily before reaching convergence, indicating stable learning. The absence of a significant divergence between the training and validation losses suggests that the model does not suffer from overfitting.

It can be seen from the Fig. 24 that data points cluster around the ground-truth prediction line. This demonstrates the high predictive capability of the proposed surrogate model while requiring only a fraction of the computational costs.

After training, the surrogate model was validated by comparing the predicted strain-stress curve to the ground-truth curve. To evaluate the predictive capability of the model, a specific beam configuration was selected from the dataset by fixing the magnetic field intensity B and the beam length L; the history of stress and strain values belonging to that fixed set were then extracted.

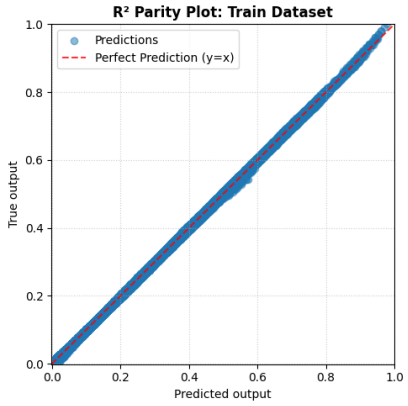
In Fig. 25 different strain-stress curves for a fixed TPU geometry and for varying B were plotted.
It can be observed that the predicted curves from the neural network reasonably approximate the real ones. The only case where there is a low correspondence to the real curve is when *B=0 T*, which represents the completely passive state of the MRF core. This discrepancy occurs because the neural network optimization path is heavily biased towards the active states, when $B > 0$, which constitutes the vast majority of the global training set.
Overall, it can be concluded that the proposed neural network accurately captures the constitutive behavior of the beam over the entire strain range.

C. Neural network for inverse design

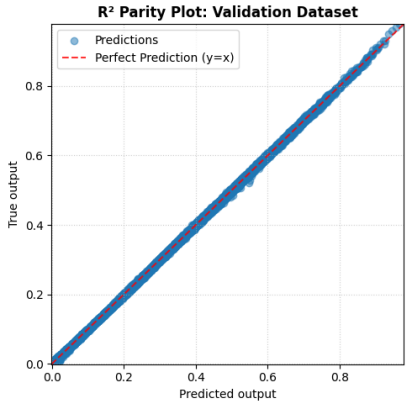
The objective is to predict as output the length of the beam L given as inputs I) the magnetic field flux B of the magnet, and ii) the apparent stiffness of the beam's mechanical response. To achieve such aim the dataset S, stated in Section VII-A, is used. The Inverse model consists of the following mapping:

$$L = f(B, log(\bar{K})). \tag{35}$$

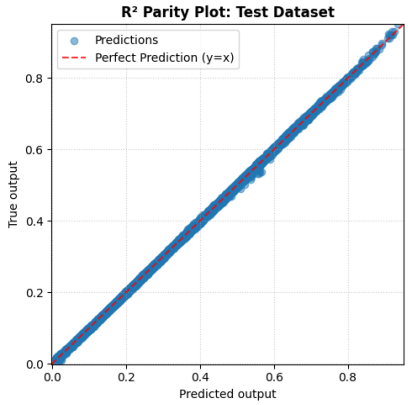
Rather than training the neural network on raw force-displacements data points, each simulation was processed to extract physically meaningful descriptors. Specifically the mean stiffness $\bar{K}$ was identified as a lumped parameter to



(a) train



(b) validation



(c) test

Fig. 22. Predicted vs estimated output normalized across all datasets: (a) training, (b) validation, and (c) test for the direct design implementation.

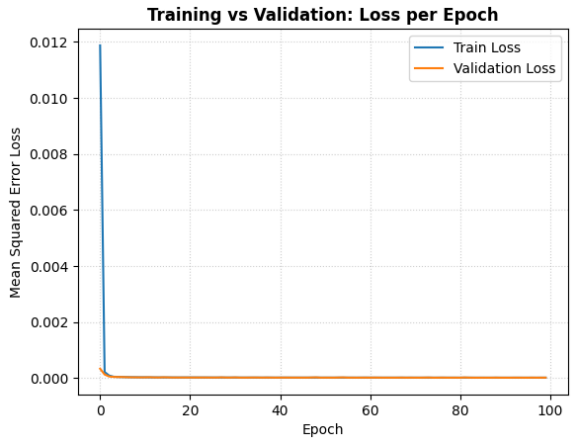


Fig. 23. Evolution trend of the training and validation losses (Direct design NNs).

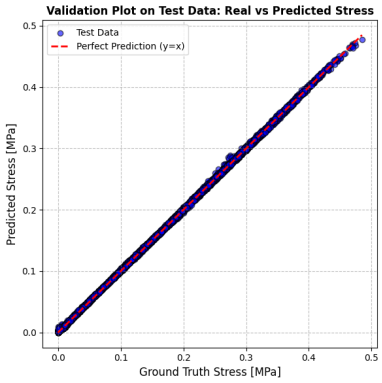


Fig. 24. Parity plot on predicted and ground truth stress data (Direct design NNs).

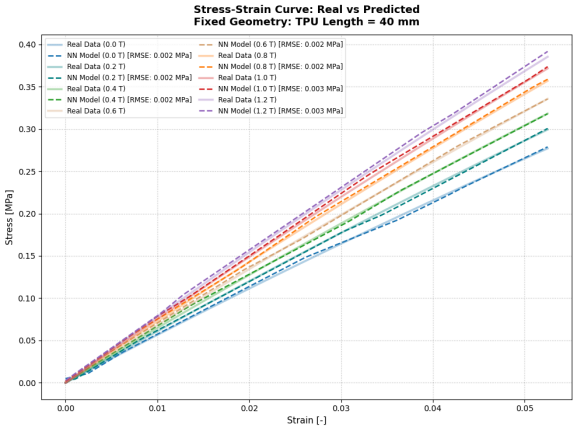


Fig. 25. Comparisons between the predicted and real stress-strain curve on different magnetic flux intensity values and fixed geometry (Direct design NNs.

characterize the overall shape of the force-displacement curve for each fixed B and L.

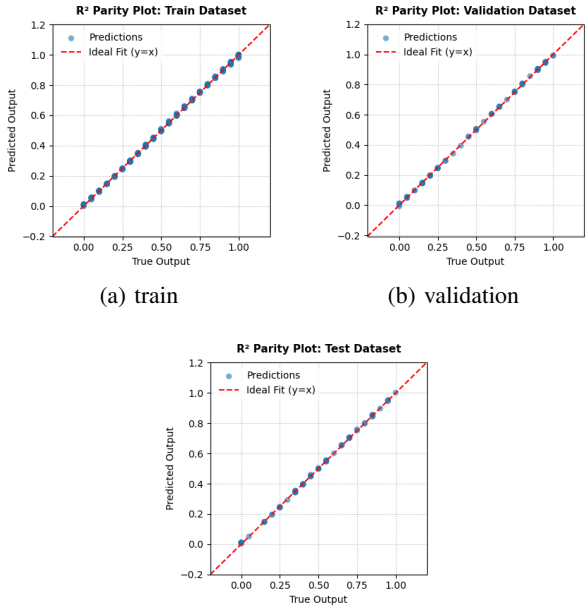
Finally, for the neural network prediction model, to improve the training stability and accommodate the potentially wide dynamic range of stiffness values, the logarithm transform of the mean of the stiffness $log(\bar{K}) \in \mathbb{R}$ is chosen as a feature, alongside with the magnetic flux density of the magnet $B \in \mathbb{R}$.

The model's network consists of an input layer with two neurons, three hidden layers with eight neurons and SiLU activation function, and an output layer of one neuron. For network training, the same loss function and training parameters from the direct model were adopted.

The Table II and Fig. 26 summarize the performance results of the inverse model. It can be observed that for the training, validation and test phases, the RMSE stay relatively low and the R^2 have high values. This means that the model is

TABLE II
PERFORMANCE RESULTS INVERSE MODEL

Dataset	RMSE	R^2
Training	0.0097	0.999
Validation	0.0102	0.999
Test	0.0101	0.999



(a) train (b) validation

(c) test

Fig. 26. Predicted vs estimated output across all datasets: (a) training, (b) validation, and (c) test results (inverse design NNs).

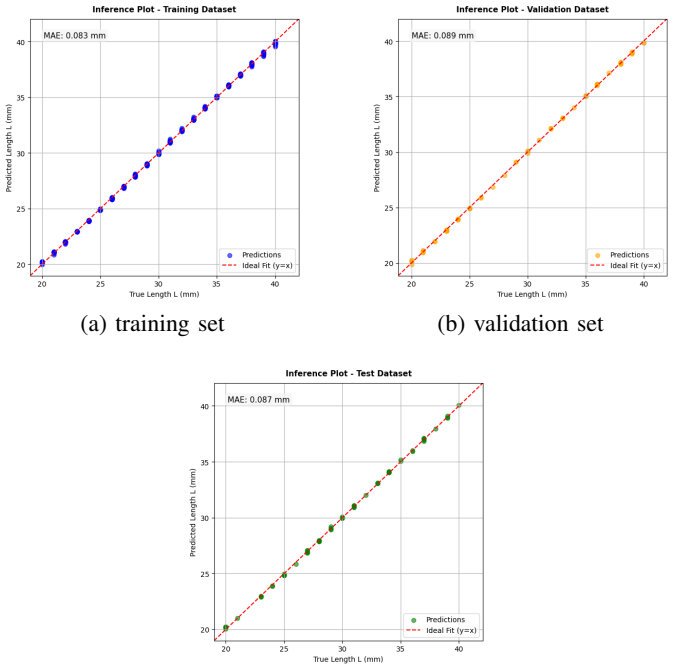
successfully learning the underlying relationship between the input and output variables.



Fig. 27. Training and validation loss for the inverse design NN.

The evolution of the training and validation losses is shown in Fig. 27. Since both training and validation follow the same trend, it can be concluded that the network is stable and it is learning properly.

Fig. 28 shows the distribution of the inference points for each length of the TPU. A smaller vertical distance from the red dashed line for a given length means that the prediction is reliable. It can be observed that not all lengths could be predicted accurately as in certain values of the TPU length the points display a wide distribution. This plot could be utilized



(a) training set (b) validation set

(c) test set

Fig. 28. Predicted vs. real output across the partitioned datasets: (a) training, (b) validation, and (c) test configurations for the inverse design network.

as a tool to determine the beam's length to be used in the experiments.

To assess the generalization and robustness of the developed inverse model an inference procedure was executed on the test dataset.

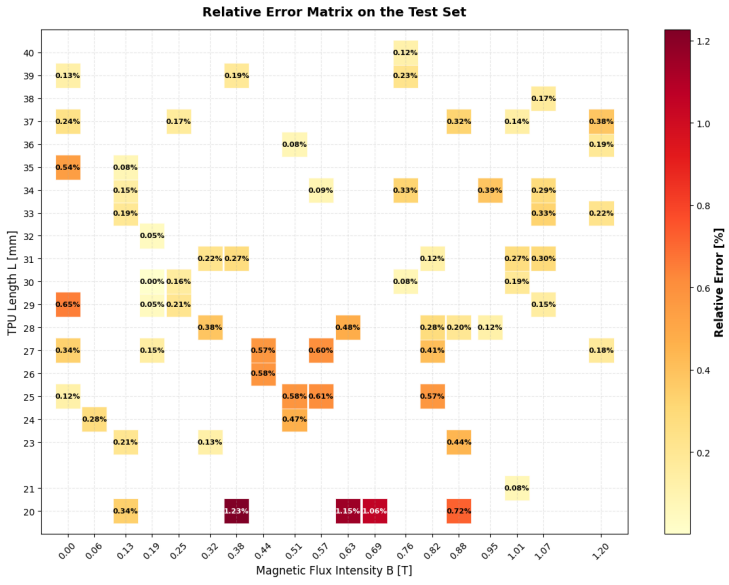


Fig. 29. Relative errors across the test set predictions (Inverse design NNs).

The spatial distribution of the relative inference error across the varying magnetic flux density and structural length is illustrated via a 2D matrix heatmap shown in Fig. 29. It can be observed from the cross-validation test plot that the relative

prediction errors are strictly bounded below 1.25%, which means that the model interpolates well on the unseen data not included in the training.

D. Neural networks for B(x) and H(x) distribution

Two simple MLP neural networks are implemented: the first learns the following function f using the dataset introduced in VII-A:

$$B(x) = f(\theta) \tag{36}$$

with $B(x) \in \mathbb{R}$, $\theta = [B, x, L] \in \mathbb{R}^3$ where B is the magnetic flux density of the magnet, x the coordinate of a point along the vertical axis of the beam of length L, it learns the magnetic flux density in that point $B(x)$ for that specific beam. The second one implements the following function f using the dataset introduced in Section VII-A:

$$H(x) = f(\theta) \tag{37}$$

with $H(x) \in \mathbb{R}$ being the magnetic field intensity at initial time of the simulation, θ being the same as above. The models can be leveraged to predict the curves $B(x)$ and $H(x)$ along the axis for a specific pair of beam length L and of magnetic flux density B at initial time. Both use a simple feedforward network consisting of an input layer of size $n_{\text{input}} = 3$, followed by two fully-connected hidden layers containing 64 and 32 neurons, respectively, each one with ReLU activation function.

The final linear output layer maps the hidden features to the target dimension $n_{\text{output}} = 1$. The network parameters were optimized by minimizing the MSE loss function, which evaluates the discrepancy between the predicted and reference field values. The optimization process is carried out using the Adam optimizer with a constant learning rate of $\eta = 10^{-3}$ (1×10^{-3}). The loss plots, as shown in Fig. 31, display a successful minimization of the loss function. The prediction on unseen data yielded satisfactory results, as illustrated in Fig. 30.

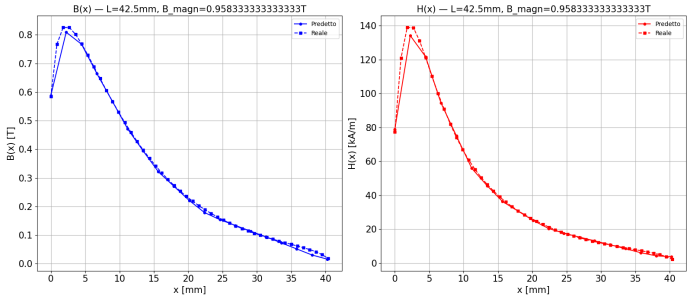
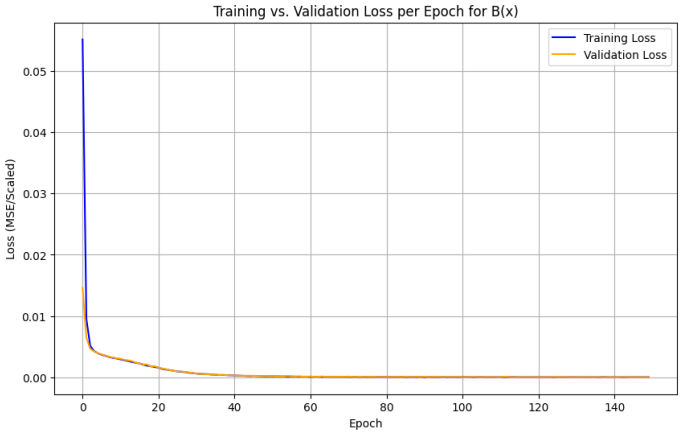
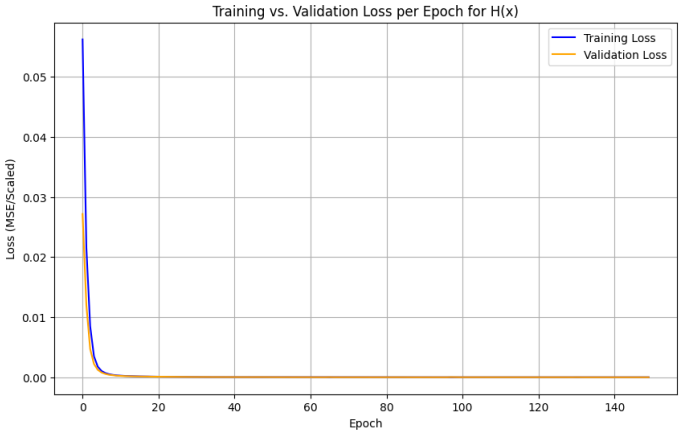


Fig. 30. Predicted curve vs true curve for new dataset values: on the left the $B(x)$ curve, on the right the $H(x)$ curve.

It is worth observing that only few points are being concentrated in the peak and the majority are distributed on the tail, thus the prediction is subjected to a larger local error where the maximum is present.



(a) loss curve for $B(x)$



(b) loss curve for $H(x)$

Fig. 31. Learning loss for (a) $B(x)$ and (b) $H(x)$.

E. Neural network for predicting ΔH from strain

The MLP learns the following function using the dataset S introduced in Section VII-A:

$$\Delta H(x) = f(B, H(x), L, \epsilon) \tag{38}$$

with $\Delta H(x) \in \mathbb{R}^m, m = 64$, $\theta = [B, H(x), L, \epsilon] \in \mathbb{R}^d, d = 67$, as B is the magnetic flux density of the magnet. Here, $H(x)$ considers the whole initial curve of 64 points along the beam axis of length L, ϵ is the beam nominal strain, $\Delta H(x)$ is the increment due to the applied strain of $H(x)$ with respect to the initial distribution, computed for each of the 64 points. A piecewise cubic Hermite interpolating polynomial interpolation procedure has been carried out to set the same number of points, m, among each simulation along the beam axis and to prevent numerical oscillations. Thus, the model can be leveraged to get from an initial $H(x)$ curve the new curve, by knowing the strain applied that achieves such profile. The architecture consists of an input layer of size d, followed by two hidden layers with 256 and 128 neurons, respectively. Non-linearity is introduced after each hidden layer using the LeakyReLU activation function. Finally, a linear output layer, provided with no activation function, maps the features to the m target dimension.

For network training parameters, an MSE loss was chosen, the learning rate was set to $\eta = 10^{-3}$, Adam optimizer was chosen, and the number of epochs was fixed at 100.

After training, the predictive performance of the neural network was assessed directly on new data through the reconstruction of the evolution of $H(x)$ curve for given strains. The results show a good generalization; the maximum error is located in the beginning of the peak where in a few points sharp changes exist, as the Fig. 32 shows:

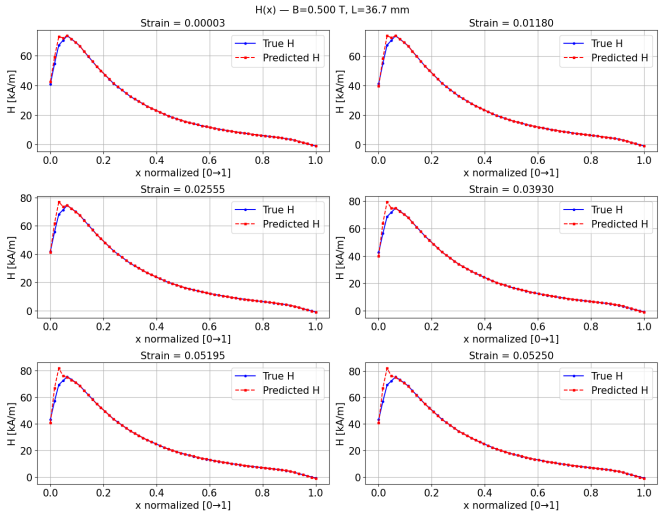


Fig. 32. Comparison of the curves for new dataset values as different strains are considered from an initial magnetic field strength curve $H(x)$.

F. Neural networks linked

Having trained the NNs separately, the new integrated model can be directly combined using the networks in Sections VII-E and VII-D, as follows:

- First network: it predicts the $B(x)$ curve based on the initial input parameters, as shown in Section VII-B.
- Second network: it takes the $B(x)$ curve as an input to generate the $H(x)$ curve, as shown in Section VII-B.
- Third network: it uses the $H(x)$ curve previously found as an input to yield the final H curve as a consequence of the applied strain, as shown in Section VII-E.

The primary objective is to create a unified workflow that predicts the entire $H(x)$ curve for a given set of input parameters and an applied strain. Consequently, the stiffness can be directly computed via Eq. (7), and the stress field follows automatically. Therefore, the model operates as a sequential pipeline where the output of each network serves as the input for the next as the trained networks are simply leveraged one after the other.

VIII. Conclusion

This work presented the development of models for actuated metamaterial consisting of a TPU beam filled with a magnetorheological fluid placed over a permanent magnet, together with a ML surrogate capable of reproducing its mechanical behavior without further FE computation.

The beam was first parametrized analytically to remain in a buckling-free, linear-elastic regime, and then progressively modeled in Comsol Multiphysics™, from a purely mechanical compression to full multiphysics coupling including the $B - H$ curve. The 2D axisymmetric formulation was shown to reproduce the 3D results while substantially reducing the computational cost, allowing the generation of the dataset used for the NN training. The comparison between the *Solid Mechanics* and the full multiphysics models, with and without a non linear magnetic permeability dependence, confirmed that the magnetic field distribution generated by the magnet is consistent across formulations.
However, the decoupled simulation strategy has proven to be not completely reliable at high values of magnet's magnetic flux density.
Several MLP-based models were trained on the datasets obtained by the FE simulations:

- a direct model predicting the stress-strain response from the magnetic field and geometry;
- an inverse model recovering the beam length from the target stiffness;
- a set of cascaded networks that reproduce the internal magnetic field distribution $H(x)$ along the beam axis and its behavior under strain.

The main limitation of the current work is that the equivalent MRF's Young's modulus formulation is based on a fully linear model without any experimental verification. Future work should perform an experimental characterization of the MRF to properly calibrate the constitutive model, then an analysis beyond the linear-elastic regime should be carried out. In fact, while the current beam was deliberately parametrized to avoid buckling and preserve a predictable linear stress-strain response, enabling the buckling phenomenon could be exploited as an additional design level. Coupled with the integration of the three cascaded networks into a single end-to-end digital shadow capable of predicting the full stress field directly from the design parameters, this would extend the applicability of the proposed framework beyond the current linear regime.

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